## Chapter 7: Algebraic methods

Chapter 7 · Original slide 1 · Algebraic methods

Pure Y1 Ch 7
Algebraic Methods

Chapter 7 · Original slide 2 · Algebraic Methods

Starter - Answers

1) Simplify:

b) $3x^2 \times 5x^5$ b) $\frac{5x^3y^2}{15x^2y^3}$

4) Factorise:

e) $x^2 - 2x - 24$

f) $3x^2 - 17x + 20$

8) Use long division to calculate:

i) $197,041 \div 23$ b) $56,168 \div 34$

11) Find the equations of the lines that pass through these pairs of points:

l) $(-1, 4)$ and $(5, -14)$

m) $(2, -6)$ and $(8, -3)$

15) Complete the square for the expressions:

p) $x^2 - 2x - 20$

q) $2x^2 + 4x + 15$

## Chapter 7: Algebraic methods

Chapter 7 · Original slide 3 · Chapter Overview

This chapter covers a number of algebraic techniques, but also algebraic ‘proofs’.

2: Dividing Polynomials

Divide $x^3 + 2x^2 - 17x + 6$

Given that $(x - 1)$ is a factor of $5x^3 - 9x^2 + 2x + a$, find the value of $a$.

4: Proof

1: Algebraic Fractions

Simplify $\frac{2x^2-5x-3}{2x^2-9x+9}$

3: The Factor Theorem

Prove that all square numbers are either a multiple of 4 or 1 more than a multiple of 4.

Chapter 7 · Original slide 4 · 1: Simplifying Algebraic Fractions

Recall that you can simplify fractions by dividing the numerator and denominator by a common factor.

$\frac{x^2-1}{x^2+x} =$

Note: Do not leave 1 in the denominator!

$\frac{2x^2+11x+12}{x^2+9x+20} =$

Tip: $\frac{a-b}{b-a} = -1$

Hint: To identify common factors we need to factorise first.

$\frac{x^2+3x+2}{x+1} =$

Tip: Factorise the easier one first because it provides clues to the factorisation of the other.

$\frac{4-x^2}{x^2+2x-8} =$

Chapter 7 · Original slide 5 · Exercise 7A – Y1 Pages 138-139

2 Simplify these fractions as far as possible:

**m** $\frac{2x^2+3x+1}{x^2-x-2}$ **n** $\frac{x^2+6x+8}{3x^2+7x+2}$ **o** $\frac{2x^2-5x-3}{2x^2-9x+9}$

3 $\frac{6x^3+3x^2-84x}{6x^2-33x+42} = \frac{ax(x+b)}{x+c}$, where $a$, $b$ and $c$ are constants.
Work out the values of $a$, $b$ and $c$.

Chapter 7 · Original slide 6 · Terminology

$11 \div 4 = 2rem3$

dividend
(the thing we're dividing)

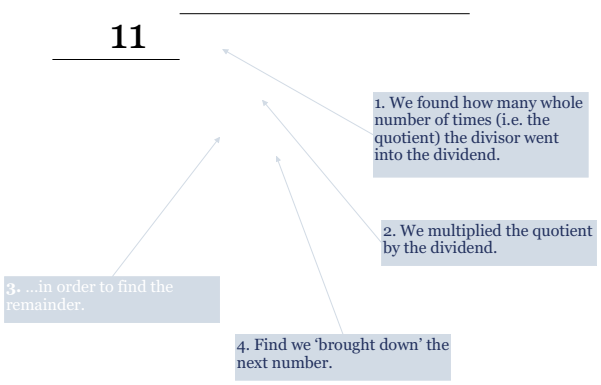
quotient

divisor
(the thing we're dividing by)

remainder

## Chapter 7: Algebraic methods

Chapter 7 · Original slide 7 · Normal Long Division




Chapter 7 · Original slide 8 · Now repeat! How many times does x go into -2{x}^{2}?

$x + 5$ | $6x^3 + 30x^2$ | $-\ 2x^2 - 7x$ | $-\ 2x^2 - 10x$ | $3x + 15$ | $3x + 15$ | $0$

Fractions

Example

How many times does $x$ go into $6x^3$?

$6x2$

$- 2x$

$+ 3$

$6x3 + 28x2 - 7x + 15$

Subtract and carry down next term.

Multiply $6x^2$ by $(x + 5)$. The first term should match with above.

Tip:

You can check your solution by expanding:

$(x + 5)\left(6x^2 - 2x + 3\right)$

But if you know you should get no remainder, ending with 0 at the bottom is a good sign!

Chapter 7 · Original slide 9 · Further Example

$x$ - 1

$3x3 + 0x2 - 2x + 4$

Find the remainder when $3x^3 - 2x + 4$ is divided by $x - 1$.

Note: Ensure that any missing terms in the polynomial are filled in, so that the powers decrease by 1 each time.

Chapter 7 · Original slide 10 · Test Your Understanding

Find the remainder when $2x^3 - 5x^2 - 16x + 10$ is divided by $x - 4$.

$x - 4$

$2x^3 - 5x^2 - 16x + 10$

Chapter 7 · Original slide 11 · Further Test Your Understanding

Divide $8x^3 - 1$ by $2x - 1$.

$8x^3 + 0x^2 + 0x - 1$

$2x - 1$

Fro Fun Fact: There is a well-known factorisation for the '*difference of two cubes*':

$x^3 - y^3 = (x - y)\left(x^2 + xy + y^2\right)$

This is NOT in the A Level syllabus.

Chapter 7 · Original slide 12 · Exercise 7B – Y1 Pages 142

15 $\mathrm{f}(x) = 10x^3 + 43x^2 - 2x - 10$.
Find the remainder when $\mathrm{f}(x)$ is divided by $(5x + 4)$.

16 $\mathrm{f}(x) = 3x^3 - 14x^2 - 47x - 14$.
a Find the remainder when $\mathrm{f}(x)$ is divided by $(x - 3)$.
b Given that $(x + 2)$ is a factor of $\mathrm{f}(x)$, factorise $\mathrm{f}(x)$ completely.

Chapter 7 · Original slide 13 · Exercise 7B – Y1 Pages 142

17 a Find the remainder when $x^3 + 6x^2 + 5x - 12$ is divided by
i $x - 2$,
ii $x + 3$.
b Hence, or otherwise, find all the solutions to the equation $x^3 + 6x^2 + 5x - 12 = 0$.

Chapter 7 · Original slide 14 · Exercise 7B – Y1 Pages 142

18 $\mathrm{f}(x) = 2x^3 + 3x^2 - 8x + 3$.
a Show that $\mathrm{f}(x) = (2x - 1)(ax^2 + bx + c)$ where $a$, $b$ and $c$ are constants to be found.
b Hence factorise $\mathrm{f}(x)$ completely.
c Write down all the real roots of the equation $\mathrm{f}(x) = 0$.

Chapter 7 · Original slide 15 · Exercise 7B – Y1 Pages 142

19 $\mathrm{f}(x) = 12x^3 + 5x^2 + 2x - 1$.
a Show that $(4x - 1)$ is a factor of $\mathrm{f}(x)$ and write $\mathrm{f}(x)$ in the form $(4x - 1)(ax^2 + bx + c)$.
b Hence, show that the equation $12x^3 + 5x^2 + 2x - 1 = 0$ has exactly 1 real solution.

Chapter 7 · Original slide 16 · 3: The Factor Theorem

$x^3 + x^2 - 4x - 4 = (x - 2)\left(x^2 + 3x + 2\right)$

We can see that $(x - 2)$ is a factor of $x^3 + x^2 - 4x - 4$.

What would happen if $x$ is 2?

The converse is also true: if we could find a value $a$ such that the LHS is 0 when we substitute in $a$ for $x$, then $(x - a)$ would be a factor.

! The Factor Theorem states that if $f(x)$ is a polynomial then:

• If $f(p) = 0$, then $(x - p)$ is a factor of $f(x)$.

• Conversely, if $(x - p)$ is a factor of $f(x)$, then $f(p) = 0$.

Chapter 7 · Original slide 17 · Examples (in textbook)

Show that $(x-2)$ is a factor of $x^3+x^2-4x-4$.

Writing $f(x)=\ldots$ gives us appropriate notation, i.e. $f(2)$, to show we're substituting 2 into the polynomial on the next line.

Keep on trying values until you find one where $f(p)=0$. Recommended order: $p=1,-1,2,-2,\ldots$

You can use the 'Table' mode on your calculator to try lots of values in a range – see my interactive Classwiz powerpoint.

Tip: We can determine b by considering for example the x terms in the expansion. This is much faster than algebraic division.

Let $f(x)=x^3+x^2-4x-4$

Fully factorise $2x^3+x^2-18x-9$.

Let $f(x)=2x^3+x^2-18x-9$

Chapter 7 · Original slide 18 · Using Factor Theorem to find unknown coefficients

Given that $2x+1$ is a factor of $6x^3+ax^2+1$, determine the value of $a$.

Notes:

Choose the value of $x$ to substitute that makes the divisor/factor 0.

So if $2x+1=0$, then $x=-\frac{1}{2}$, thus find $f\left(-\frac{1}{2}\right)$.

This even works when you're dividing by non-linear factors, e.g. $\frac{1}{x}+1$, but at A Level they will always be of the form $ax+b$.

Chapter 7 · Original slide 19 · Test Your Understanding

$\mathrm{f}(x) = 6x^3 + 13x^2 - 4$

(a) Use the remainder theorem to find the remainder when $\mathrm{f}(x)$ is divided by $(2x + 3)$. (2)
(b) Use the factor theorem to show that $(x + 2)$ is a factor of $\mathrm{f}(x)$. (2)
(c) Factorise $\mathrm{f}(x)$ completely. (4)

Given that $3x - 1$ is a factor of $3x^3 + 11x^2 + ax + 1$, determine the value of $a$.

No long in spec.

Chapter 7 · Original slide 20 · Exercise 7C – Y1 Pages 146

13 $\mathrm{f}(x) = 3x^3 - 12x^2 + 6x - 24$.
a Use the factor theorem to show that $(x - 4)$ is a factor of $\mathrm{f}(x)$.
b Hence, show that 4 is the only real root of the equation $\mathrm{f}(x) = 0$.

Chapter 7 · Original slide 21 · Exercise 7C – Y1 Pages 145-146

14 $\mathrm{f}(x) = 4x^3 + 4x^2 - 11x - 6$.
a Use the factor theorem to show that $(x + 2)$ is a factor of $\mathrm{f}(x)$.
b Factorise $\mathrm{f}(x)$ completely.

Chapter 7 · Original slide 22 · Exercise 7C – Y1 Pages 145-146

15 a Show that $(x - 2)$ is a factor of $9x^4 - 18x^3 - x^2 + 2x$.
b Hence, find four real solutions to the equation $9x^4 - 18x^3 - x^2 + 2x = 0$.

Chapter 7 · Original slide 23 · Exercise Extension

**1 2**

*[MAT 2006 1E]* The cubic

$x^3 + ax + b$

Has both $(x - 1)$ and $(x - 2)$ as factors. Determine the values of $a$ and $b$.

[MAT 2009 1I] The polynomial

$n^2x^{2n+3} - 25nx^{n+1} + 150x^7$

Has $x^2 - 1$ as a factor

A) for no values of $n$;

B) for $n = 10$ only;

C) for $n = 15$ only;

D) for $n = 10$ and $n = 15$ only.

**3**

[MAT 2013 1G] Let $n \geq 2$ be an integer and $p_n(x)$ be the polynomial

$p_n(x) = (x - 1) + (x - 2) + \ldots + (x - n)$

What is the remainder, in terms of n, when $p_n(x)$ is divided by $p_{n-1}(x)$?

The remainder theorem states that if $f(x)$ is divided by $(x - a)$, the remainder is $f(a)$. This similarly works whenever $a$ makes the divisor 0.

Chapter 7 · Original slide 24 · 4: Proof

Terminology

A conjecture is a mathematical statement that has yet to be proven.

One famous conjecture is Goldbach's Conjecture.

It states "*Every even integer greater than 2 can be expressed as the sum of two primes.*"

It has been verified up to $4 \times 10^{18}$ (that's big!); this provides evidence that it is true, but does not prove it is true!

A theorem is a mathematical statement that has been proven.

One famous misnomer was Fermat's Last Theorem, which states "*If* $n$ is an integer where $n > 2$*, then* $a^n + b^n = c^n$ *has no non-zero integer solutions for* $a, b, c$." It was 300 years until this was proven in 1995. Only then was the 'Theorem' in the name then correct!

Chapter 7 · Original slide 25 · Types of Proof

A proof must show all assumptions you are using, have a clear sequential list of steps that logically follow, and must cover all possible cases.

You should usually make a concluding statement, e.g. restating the original conjecture that you have proven.

This is the simplest type, where you start from known facts and reach the desired conclusion via deductive steps.

a. Proof by Deduction

Chapter 7 · Original slide 26 · Be Warned…

Proof by Deduction requires you to start from known facts and end up at the conclusion. It is not acceptable to start with to the conclusion, and verify it works, because you are assuming the thing you are trying to prove.

*Example*: Prove that if three consecutive integers are the sides of a right-angled triangle, they must be 3, 4 and 5.

Incorrect Proof:

Let the lengths be 3,4,5.

Therefore:

$3^2 + 4^2 = 5^2$
$25 = 25$

This satisfies Pythagoras' Theorem, and the numbers are consecutive.

Correct Proof:

We are assuming the thing we are trying to prove!

We are only assuming things in the 'if' bit. This is fine!

The underlying problem is that this technique doesn't prove there can't be other consecutive integers that work – we have only verified 3,4,5 is one such solution.

Chapter 7 · Original slide 27 · Types of Proof

a. Proof by Deduction

Prove that $x^2 + 4x + 5$ is positive for all values of $x$.

Test Your Understanding

Exam Tip: This is quite a common last parter.

Anything squared is at least 0. This is formally known as the '*trivial inequality*'.

Prove that the sum of the squares of two consecutive odd numbers is 2 more than a multiple of 8.

Chapter 7 · Original slide 28 · Exercise 7D – Y1 Pages 149-150

11 Prove that $x^2 + 8x + 20 \geq 4$ for all values of $x$.

12 The equation $kx^2 + 5kx + 3 = 0$, where $k$ is a constant, has no real roots. Prove that $k$ satisfies the inequality $0 \leq k < \frac{12}{25}$. (4 marks)

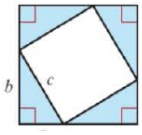
Chapter 7 · Original slide 29 · Exercise 7D – Y1 Pages 149-150

13 The equation $px^2 - 5x - 6 = 0$, where $p$ is a constant, has two distinct real roots. Prove that $p$ satisfies the inequality $p > -\frac{25}{24}$. (4 marks)

18 A circle has equation $(x - 1)^2 + y^2 = k$, where $k > 0$.
The straight line $L$ with equation $y = ax$ cuts the circle at two distinct points.
Prove that $k > \dfrac{a^2}{1 + a^2}$.

Chapter 7 · Original slide 30 · Exercise 7D – Y1 Pages 149-150

**20** The diagram shows a square and four congruent right-angled triangles.
Use the diagram to prove that $a^2 + b^2 = c^2$.




19 Prove that the line $4y - 3x + 26 = 0$ is a tangent to the circle $(x + 4)^2 + (y - 3)^2 = 100$.

Chapter 7 · Original slide 31 · Exercise Extension

*[STEP I 2005 Q1]* 47231 is a five-digit number whose digits sum to

$4 + 7 + 2 + 3 + 1 = 17.$

(i) Prove that there are 15 five-digit numbers whose digits sum to 43. You should explain your reasoning clearly.

(ii) How many five-digit numbers are there whose digits sum to 39?

Chapter 7 · Original slide 32 · Other Types of Proof

b. Proof by Exhaustion

This means breaking down the statement into all possible smaller cases, where we prove each individual case.

(This technique is sometimes known as 'case analysis')

Prove that $n^2 + n$ is even for all integers $n$.

c. Disproof by Counter-Example

While to prove a statement is true, we need to prove every possible case (potentially infinitely many!), we only need one example to disprove a statement.

This is known as a counterexample.

Disprove the statement:

"$n^2 - n + 41$ is prime for all integers $n$."

Chapter 7 · Original slide 33 · Exercise 7E – Y1 Pages 152-154

4 Prove that all cube numbers are either a multiple of 9 or 1 more or 1 less than a multiple of 9.

Chapter 7 · Original slide 34 · Exercise 7E – Y1 Pages 152-154

9 a Prove that for any positive numbers $p$ and $q$:
$p + q > \sqrt{4pq}$

b Show, by means of a counter-example, that this inequality does not hold when $p$ and $q$ are both negative. (2 marks)